

OSVALDO CESAR QUINTEROS TERRAZAS

MECHATRONICS ENGINEER / FULL-STACK WEB DEVELOPER

La Paz - Bolivia | osvaldo.quinteros.terrazas@gmail.com | +591 70135802 |

LinkedIn | GitHub

Professional Summary

Mechatronics Engineer and Full-Stack Web Developer with hands-on experience in industrial automation, software development, and data analysis. Skilled in designing and implementing solutions that integrate hardware and software, from PLC control systems to modern web applications. Strong background in robotics and aerospace research through projects such as NASA's Human Exploration Rover Challenge and Bolivia's first CubeSat mission. Passionate about building innovative, scalable, and data-driven solutions for industry 4.0 and beyond.

Education

Bachelor of Mechatronics Engineering – Universidad Católica Boliviana (2024, La Paz – Bolivia)

Relevant Coursework: Control Systems, Industrial Automation, Robotics, Programming

Diplomas & Certifications

- Diploma: Automation, Instrumentation, Intelligent Control Systems and Industrial IoT – Universidad Privada Boliviana (UPB)
- Full-Stack Web Development Bootcamp – Udemy (React, Node.js, Express, SQL)
- Python Programming – A2 Capacitaciones (Pandas, Matplotlib, Seaborn, Scikit-learn)
- Excel & Power BI – A2 Capacitaciones (Data modeling, dashboards, KPIs)
- AutoCAD Advanced – Udemy (2D/3D CAD Design)
- Mechanical Design – Dassault Systèmes (SolidWorks)
- Additive Manufacturing Associate – Dassault Systèmes (3D Printing)

Skills

- Web Development: React, Node.js, Express, PostgreSQL, REST APIs, HTML, CSS, JavaScript, Vite

- Data & Analytics: Python (Pandas, Matplotlib, Seaborn, Scikit-learn), Power BI, Excel, SQL
- Engineering & CAD: SolidWorks, AutoCAD, Additive Manufacturing (3D Printing)
- Industrial Automation: PLC programming, Robotics (KUKA KR6 R900)
- Languages: Spanish (Native), English (B2)

Experience

Maintenance Assistant – PREFORSA Plastic Packaging Industry (Jun 2024 – Oct 2024, El Alto – Bolivia)

- Developed and implemented maintenance plans for critical plant equipment, reducing unexpected downtime.
- Designed and prototyped a quality testing device for plastic packaging, improving product validation process.
- Supported root-cause analysis of machinery failures, contributing to more efficient maintenance scheduling.

Engineering Intern – Machine Learning Bolivia (Feb 2022 – Jul 2022, La Paz – Bolivia)

- Assisted in the design and manufacturing of mechatronic prototypes for industrial applications.
- Collaborated in the development of experimental setups for machine learning research projects.
- Contributed to documentation and testing of new engineering solutions.

Laboratory Assistant – Universidad Católica Boliviana (Aug 2021 – Dec 2021, La Paz – Bolivia)

- Maintained and calibrated laboratory equipment to ensure availability for student practices.
- Installed safety devices and signage to comply with laboratory standards.

- Programmed routines for KUKA KR6 R900 robotic arm, enabling student demonstrations and training.

Projects & Portfolio

- 🌐 Online Resume: Personal CV website built with HTML, CSS, and JavaScript. <https://gattosva50.github.io/resume/>
- ☀️ Weather App: React app using OpenWeather API with live weather data and responsive UI. https://gattosva50.github.io/weather_app/
- 🛒 E-commerce App (In Progress): Full-stack project with React (frontend) and Node.js/Express (backend). <https://github.com/Gattosva50/e-commerce>
- 🤖 Engineering & Research Projects: NASA HERC participation, member of Bolivia's first CubeSat mission.